



CHEMISTRY Department
SIMHADRI

Note No :1



Ref:SMTTP/Chemistry/2023-24/Approval/571710

Date:

16/10/2023(11:40:45)

Subject:LMI on flue gas based Operation & Maintenance of desalination plant

Site LMI on **Operation and Maintenance of Flue Gas Based Desalination plant** is due for review.

Following changes are done in LMI .

Section	Brief details of changes	Previous Rev & Date	Current Rev & Date
	Amendment Record added	--	Rev 2, Oct-2023
Introduction	Capacity of desalination plant corrected from 240 TPD to 120 TPD	Rev 1, Oct-2019	Rev 2, Oct-2023
Superseded document	Rev1 documents added	Rev 1, Oct-2019	Rev 2, Oct-2023
Specific instruction to be followed	LMI routing is done through all concerned depts	Rev 1, Oct-2019	Rev 2, Oct-2023

Put up for kind approval of CA for implementation of LMI.

Submitted by:

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On:16/10/2023(11:40:45) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:2

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On:25/10/2023(10:21:46) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:3

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On:25/10/2023(10:29:00) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:4

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On:25/10/2023(10:44:01) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:5

Maintenance of ID Fans, Ducts, Flue Gas Heat Exchangers etc are not in the purview of Offsite Maintenance.

May please be reviewed.

Submitted by:

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On:26/10/2023(17:47:09) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:6

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On:26/10/2023(18:12:53) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:7

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On:06/12/2023(17:06:46) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:8

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On:15/12/2023(11:29:36) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:9

As discussed please correct the Page no. 2 of 13 and also attach the schematic drawing (as built) applicable for the Simhadri plant as annexure to this newly revised LMI.

Submitted by:

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On:21/12/2023(09:58:31) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:10

As discussed

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On:21/12/2023(10:58:50) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:11

-Changes done as proposed at page:2

- Schematic drawing & Process flow diagram are added in Annexures

"LMI-Operation and Maintenance of Flue gas desalination plant 21.12.2023.docx" enclosed.

Submitted by:

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On:21/12/2023(12:08:02) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:12

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On:21/12/2023(12:09:56) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:13

Please prepare and attach a schematic of the as-built Multi-Effect De-salination plant installed at NTPC Simhadri. For e.g. The scheme in the attached revised LMI does not show even the Flue Gas circulating Fan or the inlet/outlet motorised Damper gates. Kindly re-draw and attach the scheme to match the actual as-built condition of the Multi-Effect De-salination plant at Simhadri. Also indicate there-in, all fans, pumps, valves and Dampers indicating their respective legends.

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On:21/12/2023(17:31:03) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:14

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On:21/12/2023(17:37:20) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:15

General layout of plant is attached with LMI Pl.

Submitted by:

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On:22/12/2023(13:11:20) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:16

May kindly be approved

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On:22/12/2023(17:36:57) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:17

Gullotine Gates GG1 and GG2 (see drawing placed in page no. 16 of the revised LMI) alongwith their respective sliding gates , the ID Fan exclusively shown in the Drawing attached on the page no .16 of the revised LMI and all its connected Dampers, the Heat exchanger and all other components (including all piping, valves and pumps) that go into working of the De-salination Plant **shall be maintained by Offsite Mechanical Maintenance Group**. For carrying out the above, necessary service contracts (BMC as well as Overhauling) as per the Maintenance schedule mentioned in this revised LMI or as mentioned in the O&M Manual of the De-Salination Plant shall be kept available and executed by Offsite Group. All Electrical and C&I related equipment pertaining to the entire De-salination plant shall be maintained by the respective **EMD and C&I groups** (Same area division to be followed in De-salination plant as being followed in Main Plant by both EMD and C&I groups).

The above is for information and implementation to all concerned.

For any further clarifications and queries, undersigned may please be contacted.

Submitted by:

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On:26/12/2023(13:26:39) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:18

Noted.

Submitted by:

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On:27/12/2023(19:20:48) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:19

Review

Submitted by:

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On:27/12/2023(20:06:20) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:20

All Electrical related equipments will be maintained by EMD.

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

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On:03/01/2024(16:10:57) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:21

All electrical related equipments will be taken care by EMD , however control type actuator is to be taken care by C&I as per uniform manitenance circular ED(OS).

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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On:03/01/2024(17:00:56) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:22

Pl review the LMI

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On:03/01/2024(19:28:43) **Ref:**SMTPP/Chemistry/2023-24/Approval/571710

Note No:23

C&I instruments and plc system will be looked after C&I maintenance, Motorised actuator(SW3,SW7) used in Desalination plant is inching type actuator and ON-OFF command is actuating through relays of EMD panel may please be looked after EMD.

Submitted by:

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On:04/01/2024(12:56:59) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:24

All C&I related equipments & PLC shall be maintained by C&I (except motorised actuators as per remarks of Note 23).

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On:08/01/2024(10:28:00) **Ref:**SMTTP/Chemistry/2023-24/Approval/571710

Note No:25

May please be approved.

Submitted by:

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On:08/01/2024(17:16:29) **Ref:**SMTTPP/Chemistry/2023-24/Approval/571710

Note No:26

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On:09/01/2024(06:17:32) **Ref:**SMTTPP/Chemistry/2023-24/Approval/571710

NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION

Operation and Maintenance of Flue Gas Based Desalination plant

NTPC SIMHADRI/CHEM/LMI/13

ISSUE NO: 1 Rev: 02 Date: OCTOBER-2023

LMI: Operation and Maintenance of Flue Gas Based Desalination plant	
Issue No:1	Revision No: 2
Issue Date: October-2019	Revision Date: October-2023

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AMENDMENTS RECORD

Section	Brief details of changes	Previous Rev & Date	Current Rev & Date
	Amendment record added	--	Rev 2, Oct-2023
Introduction	Capacity of desalination plant corrected from 240 TPD to 120 TPD	Rev 1, Oct-2019	Rev 2, Oct-2023
Superseded document	Rev1 documents added	Rev 1, Oct-2019	Rev 2, Oct-2023
Specific instruction to be followed	LMI routing is done through all concerned depts	Rev 1, Oct-2019	Rev 2, Oct-2023
Annexures	Process flow diagram & Schematic drawing are added	--	Rev 2, Oct-2023

1.0	INTRODUCTION
	FGSW-Desalination system is envisaged to produce De-mineralized (DM) water from sea water. The capacity of the plant is 120 TPD (Tone per day). It is envisaged to be accomplished by utilizing the thermal energy of waste flue gas. Flue gas is tapped from downstream of ID fan and sent to a heat exchanger where hot water is generated. This hot water is flashed in flash chamber to produce LP steam. Flue gas exiting heat exchanger is sent back to main flue gas duct going to chimney thru a separate flue gas fan. LP steam is taken to MED where desalinated water is produced from sea water. Desalinated water is polished to produce DM water in an electrodeionization (EDI) system. ID fan is having VFD which modulates the speed to maintain a constant hot water temp at all operating conditions. Designed Gain Output Ratio of the plant is 6 i.e with 1 T/hr steam it generates 6 T/hr desalinated water from sea water.
	MAJOR SYSTEMS AND ITS PROCESS DESCRIPTION WITH CONTROL PHILOSOPHY
	1) Sea water system 2) Brine system 3) Distillate system 4) Vacuum system 5) Flue gas heat exchanger and Flash chamber system 6) Electro deionization system 7) Dosing system
	SEA WATER SYSTEM
	The sea water is pumped to the MED (multi effect desalination) and condenser unit through sea water pumps. The sea water exiting the MED-condenser is divided into two parts. One is taken to the MED as main inlet feed and the second is sent to the ash slurry sump.
	a) Sea water return line control
	Sea water flow as measured by Sea water outlet from MED condenser flow transmitter (FT-3) will be used to control the flow by regulating the sea water return control valve (SW7).
	BRINE SYSTEM
	The MED unit will also have steam inlet from the flash chamber (described in the Flue gas and hot water system) and this steam will be used to heat the sea water. MED is a thermally driven process which utilizes the steam to produce useful distillate (described in the distillate system) and waste brine from the sea water. Brine is pumped out by a brine pump and the discharge of the brine pump is separated into two paths one for recirculation to MED (along with sea water) and the other is rejected as waste to the ash slurry sump.
	a) Brine level control
	The brine level as monitored by Brine level transmitter (LT-2) will be used to control the Brine reject line control valve (SW-14).
	b) Brine Pump VFD control
	Brine recirculation flow (FT-4) will be monitored to control the VFD of the running brine pumps

	DISTILLATE SYSTEM
	Distillate produced by the MED will be pumped by distillate pumps and will be sent to the distillate water tank (described in Electro deionization section).
	a) Condenser level Control
	The level of the condenser as measured by condenser level transmitter (LT-3) will be monitored to control the Distillate pump discharge control valve (DS-7).
	VACUUM SYSTEM
	Vacuum pumps are used to create vacuum in the MED for initial start-up and then to remove non condensable gas from the MED system during operation.
	FLUE GAS HEAT EXCHANGER AND FLASH CHAMBER SYSTEM
	Small portion of flue gas (75 – 125 TPH) is drawn from existing ID fan discharge by a VFD driven Flue gas fan and is taken to a Heat exchanger. The flue gas exiting the Heat exchanger will be sent back to the main duct.
	a) ID fan (M05) VFD control
	DM water outlet temperature from the heat exchanger (TT-4) is monitored and is used to vary the speed of the ID Fan through the VFD. The outlet temperature should be maintained at a constant temperature of 90Deg C.
	b) Flash Chamber System (0.34 bar & 72°C).
	DM water heated inside the heat exchanger is taken to a Flash chamber where it is flashed under vacuum to produce low pressure saturated steam. This steam is sent to the MED (described in Brine system). 2 - 3% of the hot water entering flash chamber is converted to steam and the remaining in liquid form is pumped back to the heat exchanger through hot water pumps.
	ELECTRO DEIONIZATION SYSTEM (EDI)
	Distillate from the distillate pump is stored in a distillate tank. EDI pumps are used to pump the distillate from the distillate tank for the EDI process. After the EDI process is complete the discharge is pumped by DM water pumps to the existing DM water tank in Stage-1.
	EDI inlet parameters: Conductivity = 10 mic-µs/cm (Max) TDS = 5ppm (Max), Pr = 6 bar (Max), Temp = 50°C (Max)
	DOSING SYSTEM
	Three dosing systems are envisaged in the process and will have each two pumps and a level transmitter.
	1). Anti scalant dosing system, 2). Anti foam system and 3) dosing system for maintaining pH

PLANT INPUT & OUTPUT DATA			
	SI No	Description of the system	Design conditions
	1	Quality of desalinated water	Conductivity = 10 mic-µs/cm (Max)
	2	Quality of DM water	Conductivity = 0.1 mic-µs/cm (Max) Silica = 15ppb (Max) PH = 6.5-7.5
	3	Quality of inlet sea water used	TDS = 45000-60000ppm PH = 7.6-8.0 Temp = 32.5°C
	4	Quality of Flue Gas (Gas composition) used	O2 = 6-7% CO2 = 12-13% SOx = 1300-1600 mg/Nm³ NOx = 400-500 mg/Nm³ SPM = 200-400 mg/Nm³ Temp =125°C-130°C
	5	Guaranteed parameters of FGSW-Desal system	Generation capacity of distillate =6 T/hr at maximum hot water flow of 45 Ton/hr & maximum hot water temp of 90°C.
2.0	SUPERSEDED DOCUMENT		
	Issue:1 Rev:1		
3.0	REFERENCE DOCUMENT		
	Technical specifications / Documents provided by OEM& NETRA.		
4.0	SCOPE		
	The scope of this LMI includes the Flue Gas Based Sea Water Desalination plant provided at Unit-1 (500MW) of NTPC-SIMHADRI.		
5.0	SPECIFIC INSTRUCTIONS TO BE FOLLOWED		
S No	Activity	Responsibility	
5A	GENERAL MONITORING AND CARE DURING PLANT OPERATION		
1	All running equipment shall be checked for proper lubrication, vibrations, abnormal noises, leakages, overheating of bearing, annunciations etc.on a continuous basis.		HOD (Chemistry)
2	In each shift a log sheet form will be filled in with the operating data collected at control system panel and on local instruments.		

3	Once in a shift , samples of seawater, distillate and brine blow-down will be collected and analyzed at chemical laboratory for determining conductivity and pH values. Any anomalous value or trend shall be recognized and reported to the responsible Engineer.	
4	The emergency shutdown operation is in a way a quick form of normal shutdown. The first action required is to reduce or stop the steam supply to the MED in order to avoid overshooting of the brine temperature.	HOD (Chemistry)
5	The sizing of Sea Water Pump is done conservatively to that even with sea water temperature of 34°C(TT-7) at MED-Condenser inlet, the outlet sea water temperature does not exceed 45°C (TT-8). This shall allow the MED-Condenser back pressure to be held at or below 106 mbar (PT-4) i.e a saturation temperature of 47°C (TT-8) under all operating conditions.	
6	Monitor the quantum of re-circulated brine is measured through a flow transmitter (FT-4) and controlled with a control valve (SW-14) in brine reject line in a manner that the maximum salt concentration at MED outlet is equal to or less than 7%.	
7	Monitor the quantum of sea water exiting MED-Condenser. Flow towards MED is controlled with a control valve (SW-7) in sea water return line in a manner so that the total flow (brine re-circulation + MED Condenser to MED sea water flow) entering MED is held constant at 32 Tons/Hr at all operating conditions. Measurement to total flow shall be done through a flow transmitter (FT-3).	
8	Never allow the flue gases to flow through the Flue Gas Heat Exchanger when the Hot water circulation through the FGHE is not on.	
9	Opening a damper of a working Heater manually when water flow is not taking place through the FGHE can pose danger and equipment damage. Hence this option is to be used very carefully by qualified and experienced personnel only. Sufficient care may be taken to ensure that the water flow through the FGHE is well established before using this provision.	
10	In case any maintenance is required on the FGHE, then close the DM water inlet / outlet valves and close Flue gas dampers at FGHE inlet & outlet before starting work.	
11	Never operate the system at pressures and / or temperatures above the design operating values	
12	The temperature of motive steam to evaporator shall not exceed 64.5°C under normal operation. The above top brine temperature cannot be exceeded because otherwise the possibility of scaling formation would increase and the effect of the anti-scale additive would be lost. High temperatures can be caused by excessive steam to the thermo-compressor or excessive fouling in the heat transfer tubes.	HOD (Chemistry)

S No	Activity	Responsibility
5B	MAINTENANCE ASPECTS	
1	Ensuring Mechanical isolation of equipment before maintenance of drives	HOD(Chem)
2	Ensuring electrical isolation of equipment before maintenance of drives	HOD (Operation)
3	Flue gas heat exchanger is to be tested hydraulically / pneumatically once in a year.	HOD (Offsite MM)
4	In case fouling of heat transfer areas internally due to deposits or if hard scales are formed on the waterside, acid cleaning to be carried out.	

5	The Desalination Plant should be shutdown at intervals of no more that 12 months running time. During the shutdown period all programmed and necessary special maintenance activities are to be carried out.	
6	Schedule of Maintenance to be followed	
1	Chemical cleaning of MED with Formic Acid. All the damaged tubes Ti / Al alloy inside MED to be replaced	Yearly once.
2	Inspection of flash chamber. If not found Ok then clean with Tri Sodium Phosphate.	Yearly once
3	All rotating drives	once in 12 weeks
4	Cleaning of FGHX with DM water	once in 3 months
5	Inspection of FG gates & chain mechanism	Yearly once
6	Flue gas duct inspection	Once in 2 years
7	FGHX fins thickness measurement	Once in 2 years
8	Inspection of Demistifiers and cleaning inside MED	Yearly once
9	ID fan Preventive maintenance	Once in 3 months
10	FEDI inspection	Yearly once
11	Final condenser tubes inspection	Yearly once
12	All chromets / gaskets inspection and replacement	Yearly once
13	All Electrical drives preventive maintenance	Once in 6 Months
14	All Electrical drives Servicing	Yearly once
15	All Electrical actuators Servicing	Yearly once
16	Electrical SWGR servicing	Yearly once
17	PLC system preventive checks	Yearly once

HOD(EM)

HOD(C&I)

S No	Activity	Responsibility
5C	Plant Preservation	
1	Washing: Drain and flush with fresh water the shell side of the Multi-Effect Evaporator and associated piping and equipment after plant shut-down, in order to protect the equipment from the pitting corrosion associated to the presence of pockets of stagnant, high-chloride content solution. The purpose of the operations described here is to reduce the chloride concentration inside the unit to not more than 500 mg/l. The fresh water to be used for flushing will be distillate water, or anyhow water having a TDS value less than 30 ppm and pH greater than 6.	HOD (Chemistry)
2	Outage period shorter than 24 hours: For a short term shut-down period (not longer than 24 hours), keep the plant under vacuum conditions.	

3	Outage period between 24 and 72 hours: After completing the washing and draining of the Evaporator, the plant should be kept under vacuum conditions, using the vacuum system.
4	Outage period above 72 hours: The most important factor in plant preservation is the washing with fresh water to remove salts down to a concentration of not more than 1000 $\mu\text{S}/\text{cm}^2$. This is best achieved by draining the plant off brine as completely as possible, then filling the Evaporator with fresh water and circulating it through the evaporator to ensure that all surfaces are washed.
5D	Do's And Don'ts
1	Never allow the flue gases to flow through the HRU when the Hot water circulation through the HRU is not on.
2	Opening a damper of a working Heater manually when water flow is not taking place through the HRU can pose danger and equipment damage. Hence this option is to be used very carefully by qualified and experienced personnel only. Sufficient care may be taken to ensure that the water flow through the HRU is well established before using this provision.
3	In case any maintenance is required on the HRU, then ensure isolation of ID fan suction and discharge dampers and Hot water circulation pumps & valves before starting the work.
4	The system water to be tested periodically to confirm its fitness to use. Initially the tests are required to be done once in 3 weeks and later on once in 3 months after the parameters are stabilized. The test and necessary correction are required after each major filling up or change of system water on the MED side.
5	Never operate the system at pressures and / or temperatures above the design operating values.
6	All important parameters are to be recorded in the log sheet as per the annexure
6.0	Review
	Head of Chemistry is responsible for review of this document on a 2 yearly basis and as & when required.

Annexures

Table 1: Performance Data Table for the Flue Gas Heat Exchanger (FGHX) Circuit

DATE	TIME (HRS)	Flue Gas Inlet Temperature (TT1, °C)	Flue Gas Outlet Temperature (TT2, °C)	Differential Pressure in Flue Gas (DPT1, bar)	Cold Water Inlet Temperature (TT5, °C)	Hot Water Outlet Temperature (TT4, °C)	Cold Water Inlet Pressure (PT1, bar)

Table 2: Performance Data Table for the Flue Gas Heat Exchanger (FGHX) Circuit

DATE	TIME (HRS)	Cold Water Inlet Flow Rate (FT1, TPH)	Cold Water PH (PH1)	Flash Chamber Outlet Steam Temperature (TT6, °C)	Flash Chamber Outlet Steam Pressure (PT2, bar)	Flash Chamber Water Level (LT1, mm)

Table 3: Performance Data Table for the Multi Effect Distillation (MED) Circuit

DATE	TIME (HRS)	Inlet Sea Water Flow (FT2, TPH)	Inlet Sea Water Conductivity (C1, µS/cm)	Inlet Sea Water Pressure (PT3, bar)	Inlet Sea Water Temperature (TT7, °C)	Sea Water recirculation Temperature (TT8, °C)	Sea Water recirculation Flow (FT3, TPH)	Sea Water recirculation Conductivity (C2, µS/cm)

Table 4: Performance Data Table for the Multi Effect Distillation (MED) Circuit

DATE	TIME (HRS)	mixing of Sea Water and brine Temperature (TT9, °C)	MED effect -1 Temperature (TT3, °C)	Brine Water Level (LT2, mm)	Brine Water recirculation flow (FT4,TPH)	Brine Water recirculation Pressure (PT5 bar)	Brine Water recirculation Conductivity (C3, µS/cm)	Brine recirculation Temperature (TT10, °C)

Table 5: Performance Data Table for the Multi Effect Distillation (MED) Circuit

DATE	TIME (HRS)	Distillate Water Level (LT3, mm)	Pressure of Final Condenser (PT4, bar)	Distillate Water pump outlet Pressure (PT6, Bar)	Distillate Water outlet Flow (FT5, TPH)	Distillate Water outlet Conductivity (C4, µS/cm)

Table 6: Performance Data Table for the Electro De-Ionization (EDI) Circuit

DATE	TIME (HRS)	Distillate Tank Water Level (LT4 , mm)	Distillate Tank Outlet Pressure (PT7, bar)	DM Water Inlet Silica Content (SI1)	DM Water Inlet PH (PH2)

Operation and Maintenance Flue Gas Based Sea Water Desalination plant

Table 7: Performance Data Table for the Electro De-Ionization (EDI) Circuit

DATE	TIME (HRS)	DM Water Inlet Conductivity (C5, $\mu\text{S}/\text{cm}$)	PT-8 DM Water Inlet Pressure (PT8, bar)	DM Water Outlet Pressure (PT-9, bar)	DM Water Outlet Flow (FT6, TPH)

Schematic drawing of Flue gas-based desalination plant at Simhadri

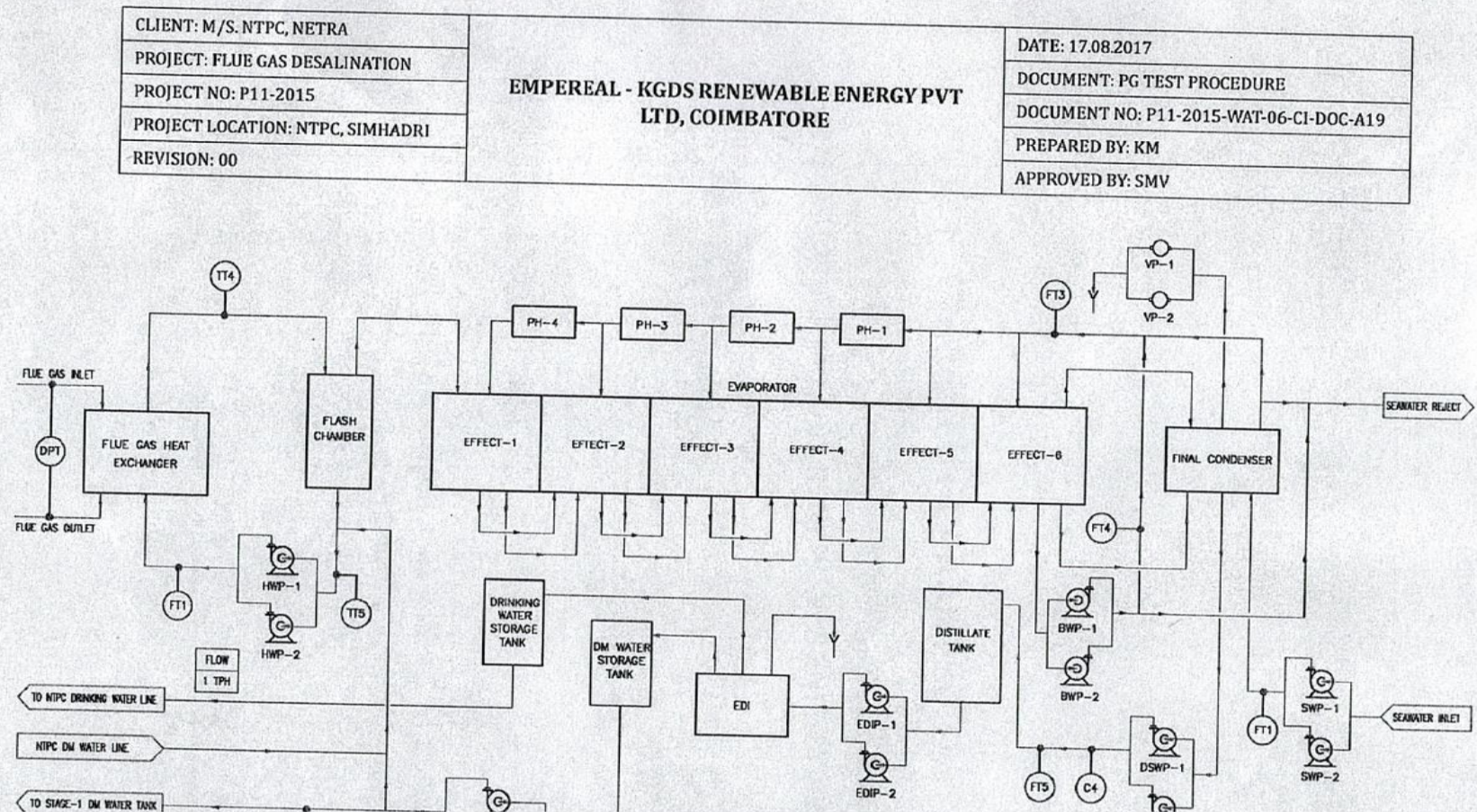
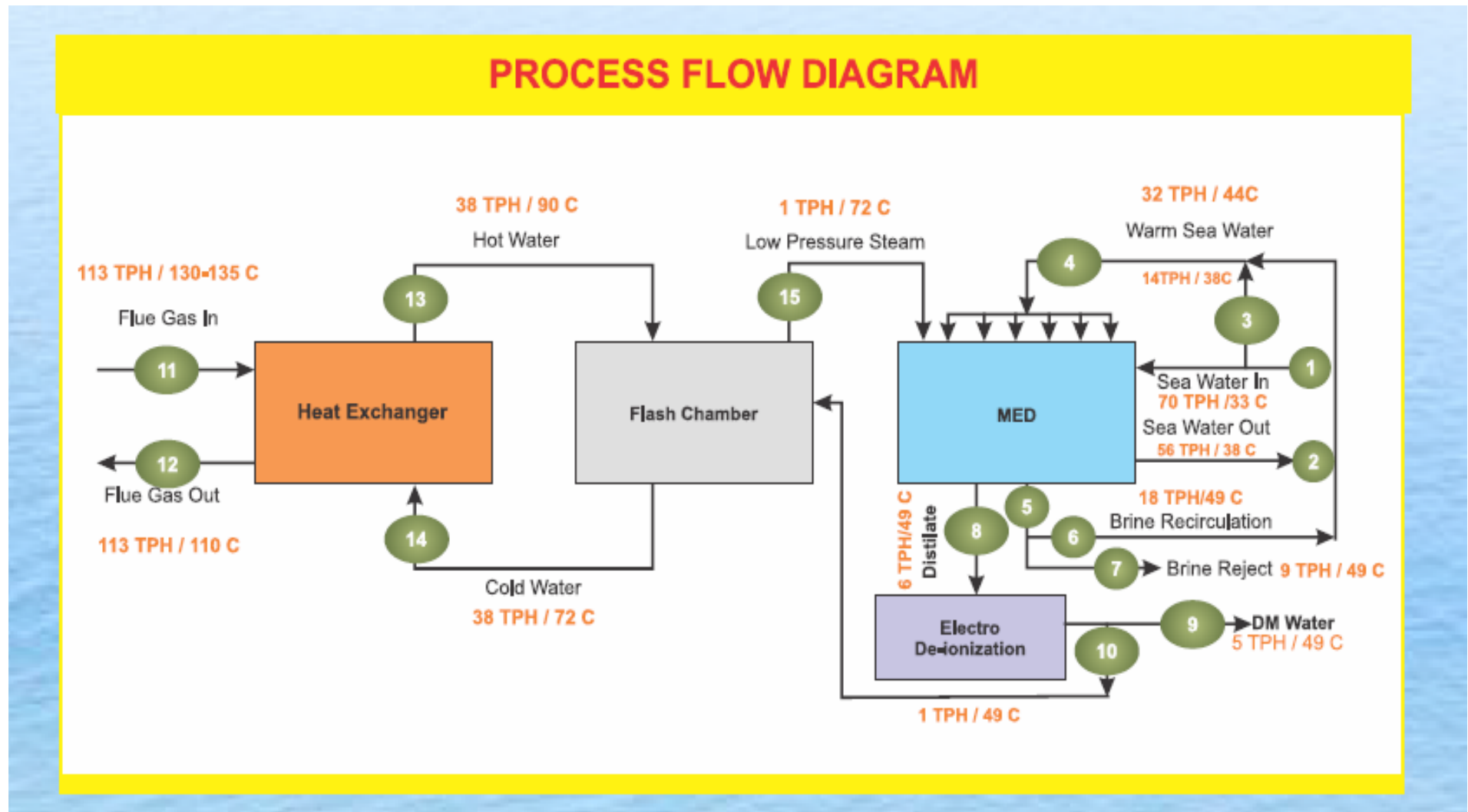
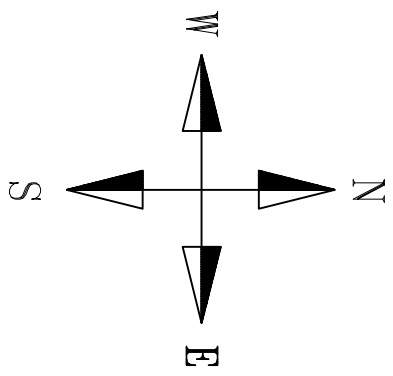


Figure 1: Flue Gas based Sea Water Desalination System (FGSW-Desal)

Process Flow Diagram





REFERENCE DRAWINGS:

Sl. NO.	
01	LAYOUT OF FLU
02	PLANT GENERA

REV.	ZONE	DESCRIPTION	DRAWN	APPD.	DATE
01	-	ROOFED AS PER MPE COMMENTS.	K.M	S.S	28.05.16
02	-	ROOF, DISTILLATE, VACUUM PUMP AND CONTROL ROOM LOCATION ROOFED.	K.M	SRI	23.10.16
03	-	01.2M WATER PUMP, DISTILLATE WATER TANK & CHEMICAL DOSING SUMP LOCATION ROOFED.	J.SS	CSK	28.05.16
04	-	02. 1M WATER TANK AND CONTROL ROOM DETAILS ROOFED.	J.SS	CSK	28.05.16

CUSTOMER:		M/S. NTPC, SIMHADRI			
FOR CUSTOMER	✓	FOR APPROVAL	✓	FOR PLANNING	✓
FOR PRODUCTION	✓	FOR ERECTION	✓		
  SCALE NTS DATE 14.12.15 14.12.15 14.12.15 14.12.15 SHEET AI TITLE : PLANT PLOT PLAN					
DRAWING No.: P11-2015-WM1-06-ME-DWG-3 PROJECT P11-2015-WM1-06-ME-DWG-3					
DRAWING No.: P11-2015-WM1-06-ME-DWG-3 PROJECT P11-2015-WM1-06-ME-DWG-3					